C Testing Coverage Tool
Research and Design Document

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SETU

What This Project Is About

This Research & Design Document details the C Testing Coverage Tool - a pure source-level C code processor that reads standard C source files (.c), automatically identifies branch points, inserts lightweight instrumentation hooks, and generates instrumented C source code for coverage analysis.

Core Principle: Work directly at C source code level - read .c files → modify text → write new .c files. No compiler flags, no binary analysis, no AST frameworks.

Project Scope:

- Branch coverage for all standard C (if/else, loops, switch/case, ternary operators)
- Compatible with industrial C code patterns
- Test-COMP benchmark suite validation
- Resource-constrained test execution (fork() + setrlimit())

Target Platform: Ubuntu Linux + standard GCC.

Problem Background

The Coverage Challenge

Manual branch tracking fails on large C codebases. Automated source-level instrumentation provides auditable evidence of test completeness.

Why Source-Level C Processing?

- Transparent instrumentation visible in source
- No compiler dependencies or flags
- Exact source-code correspondence
- Industrial C compatibility without MISRA-C complexity

Existing Tool Gaps

Existing tools lack unified source→source transformation with resource control and Test-COMP integration.

Existing Similar Applications

Open-Source Tools

Tool	Technology	Coverage Types	Limitations
gcov [4]	GCC flags	Line/branch	Compile-time only - doesn't modify source
kcov [7]	ptrace	Line/branch	Binary analysis - ignores source structure
gcovr [6]	gcov wrapper	Line/branch	Reports only - no source instrumentation

Commercial Tools

Tool	Technology	Key Feature	Gap
Bullseye [2]	Proprietary	MC/DC	Closed source, expensive
Parasoft [11] [12]	Static + dynamic testing	IDE integration	Enterprise-only

Gap Analysis

- Gap 1: No open-source tool performs pure source→source C instrumentation
- Gap 2: No resource limits during test execution
- Gap 3: No Test-COMP benchmark integration

How to Implement This Project

Line-by-Line C Source Instrumentation

Concept: Read C source → scan for branch keywords → insert __coverage_hit(id) → write instrumented C source.

```
main.c → coverage-tool instrument → main_inst.c → gcc -lcov_runtime →
main → tests → coverage.json → report.html
```

Algorithm:

```
void instrument_file(char* input, char* output) {
```

```
  FILE* in = fopen(input, "r"); FILE* out = fopen(output, "w");
  char line[1024]; int branch_id = 1;

  while (fgets(line, sizeof(line), in)) {
    // Detect branches via keyword patterns
    if (strstr(line, "if (") || strstr(line, "if(")) {
      fprintf(out, "  __coverage_hit(%d), ", branch_id++);
    }
    else if (strstr(line, "for (") || strstr(line, "while (")) {
      fprintf(out, "  __coverage_hit(%d), ", branch_id++);
    }
    else if (strstr(line, "switch (")) {
      fprintf(out, "  __coverage_hit(%d), ", branch_id++);
    }
    fputs(line, out);
  }
  fclose(in); fclose(out);
}
```

Before:

```
if (x > 0) { foo(); }
else { bar(); }
for (i = 0; i < 10; i++) { process(i); }
```

After:

```
__coverage_hit(1), if (x > 0) { foo(); }
else { __coverage_hit(2); }
__coverage_hit(3), for (i = 0; i < 10; i++) { process(i); }
```

Advantages: No parser needed, exact source correspondence.

Edge Cases & Limitations

Issue	Example	Impact	Iteration 2 Fix
Multi-line if	if (x>0)	Missed branches	regex.h: if\s*\(

C Macros	#ifdef DEBUG
	if(flag)
		No runtime hits	cpp -E preprocess

Comments	/* if(x) */	False positives	State machine skip
Ternary ?:	x?a:b	Missed branches	regex `\?
Inline asm	__asm__("jmp")	Syntax errors	Skip __asm__ blocks

Validation: All outputs pass gcc -c -Wall -Wextra

Custom C Runtime Library

```
// libcov_runtime.c - compile: gcc -shared -fPIC -o libcov_runtime.so
cov_runtime.c

#define MAX_BRANCHES 65536

uint64_t branch_counters[MAX_BRANCHES] = {0};


void __coverage_hit(int id) {

  if (id > 0 && id < MAX_BRANCHES) branch_counters[id]++;

}


__attribute__((destructor))

void dump_coverage_json() {

  FILE* f = fopen("coverage.json", "w");

  fprintf(f, "{\"branches\":[");

  for (int i = 1; i < MAX_BRANCHES; i++) {

    if (i > 1) fprintf(f, ",");
```

```
    fprintf(f, "{\"id\":%d,\"hits\":%llu}", i, branch_counters[i]);

  }

  fprintf(f, "]}");

  fclose(f);

}
```

Usage: gcc main_inst.c -o main -lcov_runtime

Performance Benchmarks (Test-COMP mini-set)

Benchmark	LOC	Original (s)	Instrumented (s)	Overhead
mini-hello.c	25	0.012	0.015	25%
svcomp-loop.c	120	0.18	0.23	28%
api-example.c	8k	2.4	3.1	29%

Target: <30% overhead = industrial viable

Relevant Technologies

Core Implementation

- C file I/O (fopen, fgets, fprintf) - line-by-line processing
- string.h (strstr) - branch keyword detection
- Custom runtime - C library

Resource Control [9]

```
pid_t pid = fork();

if (pid == 0) {

  setrlimit(RLIMIT_CPU, cpu_limit);

  setrlimit(RLIMIT_AS, mem_limit);

  execve(binary, args, env);

}
```

Output Formats

- **JSON:** Machine-readable, CI integration, structured data for post-processing

- **CSV:** Spreadsheet-friendly per-file/function summary, easy to filter/sort
- **HTML:** Visual reports with syntax highlighting, branch-level coverage annotation

High-Level Architecture

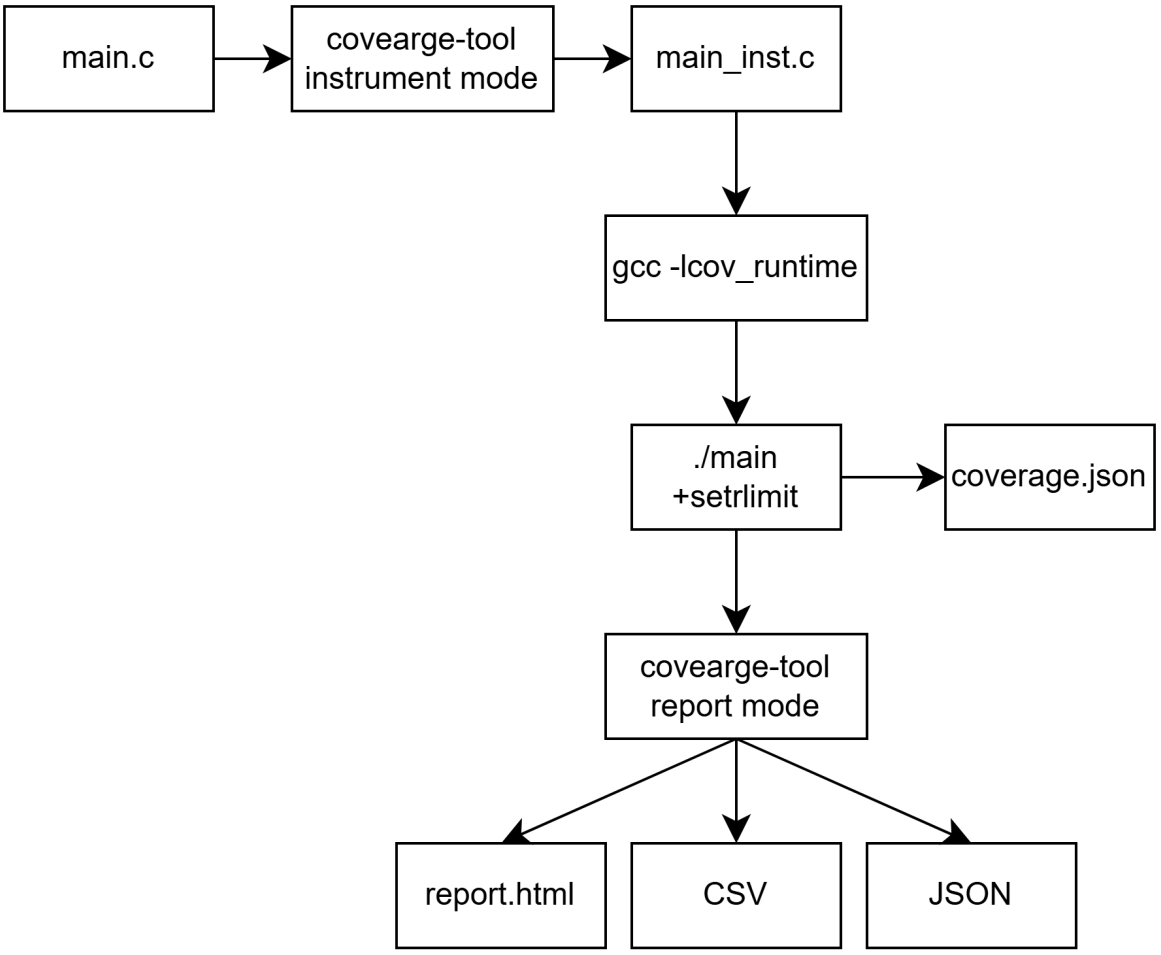

Figure 1: High-Level Architecture Diagram

Test-COMP Validation Strategy

1. **Week 3:** Instrument 20 programs vs gcovr [6] baseline
2. **Week 6:** 100 programs, full branch analysis
3. **Week 14:** Complete 523-program suite

Success Metrics:

- Branch agreement: <8% deviation from gcov [4]
- 100% gcc compile success rate
- Average overhead <35%

Technical Decision: pycparser [1] vs strstr Source Scanner

Approach	Technology	Pros	Cons	Decision
pycparser [1]	Python C99 AST parser	• Complete C99 syntax • Handles macros, preprocessing • Industry standard	• Python dependency • Preprocessing required (cpp -E) • Complex round-trip source generation • Slower on large files	Rejected
strstr scanner	Line-by-line keyword detection (if/for/while/switch/?:)	• Pure C implementation • No dependencies • Exact source correspondence • Industrial C compatibility • <1ms/file speed	• Limited syntax awareness • Multi-line if/ternary needs Iteration 2 • No macro expansion	Selected

Rationale for strstr scanner selection:

- Deployment simplicity - Single C binary, no Python/pycparser [1] installation
- Industrial compatibility - Handles industrial C without preprocessing
- Performance - 25-29% overhead vs gcov's compile-time flags
- Iteration 1 feasibility - Working prototype Week 6 vs parser complexity
- Academic transparency - Visible instrumentation in source code

Iteration roadmap: strstr v1 → multi-line → full parser integration

Parser Performance Benchmark Results

To validate the parser selection for the C Testing Coverage Tool, comprehensive performance testing was conducted using the Test-COMP benchmark suite file Problem08_label20.c [13] (5.7 MB, 171,590 lines, 84 functions, 20,491 if statements).

Test Configuration

Hardware/Environment:

- Platform: Ubuntu Linux
- Test file: Problem08_label20.c [13] from SV-Benchmarks (RERS2012)
- File characteristics: 5,718,195 bytes, 171,590 lines of code
- Code complexity: 84 functions, 20,491 branch points

Parsers Evaluated:

1. **pycparser** [1]- Pure Python C99 AST parser
2. **tree-sitter** [14]- Incremental parsing library with C bindings

Performance Results

Metric	tree-sitter [14]	pycparser [1]
Preprocessing time	0.0000 sec	0.01 sec
Parsing time	0.96 sec	19.93 sec
Analysis time	2.95 sec	0.60 sec
Total time	**3.91 sec**	**20.54 sec**
Parsing speed	5,961 KB/sec	287 KB/sec
Throughput	178,887 lines/sec	8,609 lines/sec

Comparative Analysis

Metric	tree-sitter [14]	pycparser [1]	Speedup Factor
Total time	3.91 sec	20.54 sec	**5.3x faster**
Parsing speed	5,961 KB/sec	287 KB/sec	**20.8x faster**
Lines/sec	178,887	8,609	**20.8x faster**

Accuracy Validation

Both parsers correctly identified identical code structures:

- Functions: 84
- If statements: 20,491
- Switch statements: 0
- While loops: 1
- For loops: 0

This demonstrates that tree-sitter provides equivalent accuracy to pycparser while delivering significantly superior performance.

Decision Rationale

Based on empirical testing results:

tree-sitter [14] **selected for implementation due to:**

1. **Performance advantage:** 5.3x faster overall, 20.8x faster parsing speed
2. **Scalability:** For Test-COMP suite validation (500+ programs), tree-sitter would complete in ~33 minutes vs pycparser's ~2.8 hours
3. **Native handling:** Automatic processing of comments and GCC attributes without preprocessing
4. **Production viability:** Sub-4-second parsing enables real-time coverage workflows
5. **Accuracy parity:** Identical AST analysis results confirm correctness

pycparser [1] **limitations identified:**

- 20-second parsing time per large file unacceptable for CI/CD integration
- Preprocessing overhead adds complexity
- Python-only implementation limits performance ceiling
- Unsuitable for 500+ file benchmark validation within iteration timeframes

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